AD

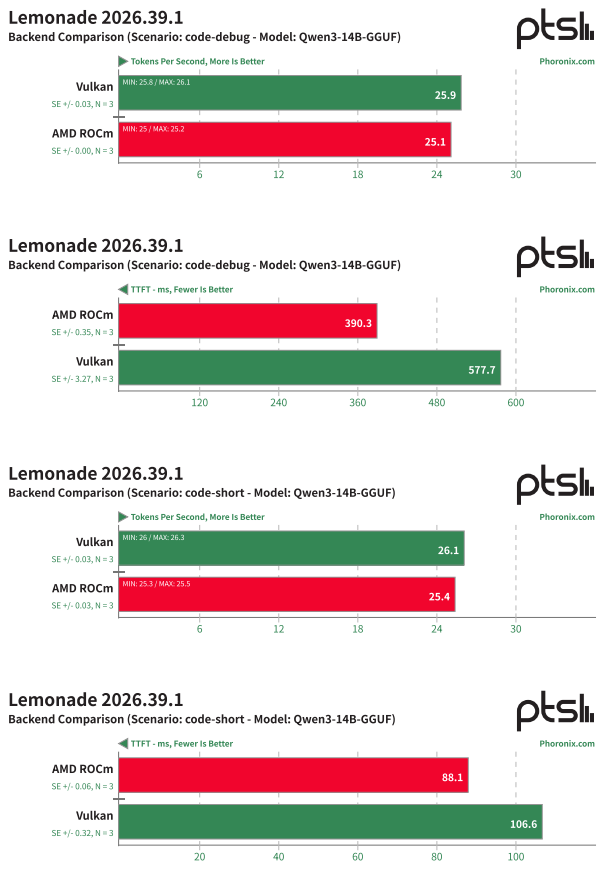
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with

AMD Ryzen AI Max+ 395 Strix Halo.



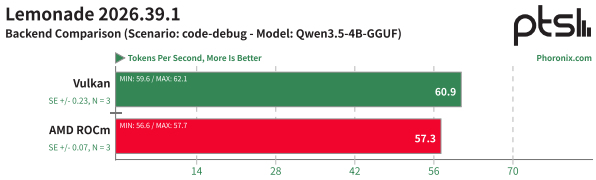
We see right away that Vulkan was delivering more tokens-per-second with Qwen3 14B compared to ROCm but the

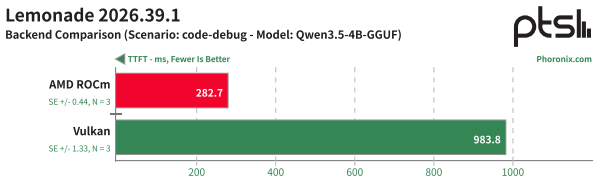
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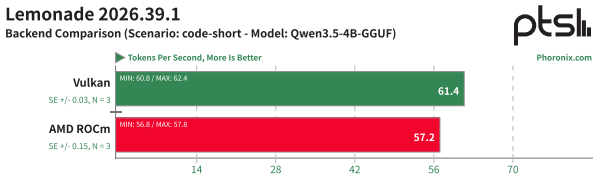
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- Intel ISPC 1.32 Released With AVX10.2 & APX Support, Even AMD Zen 6 Too
- Linux Kernel Developers Plan To Remove "Assisted-By" Tag Requirement For AI/LLM Usage
- Firefox 159 Beta Released With Fix For Tabs On Wayland
- Linux Mint 27 Continues Working Toward Christmas Release With Many Features
- ONLYOFFICE 10.0 Released With AI Features, 1,000+ Bug Fixes
- NVIDIA's Nova Driver Continues Leading The Way For Linux Rust GPU Drivers
- Asterinas 0.19.0 Brings Up X.Org & Other Improvements To This Rust-Written OS
- FreeType 2.15 Released With Apple HVF Driver, ISO Open Font Format Edition 5
- Intel Announces The GPU D Series For Crescent Island

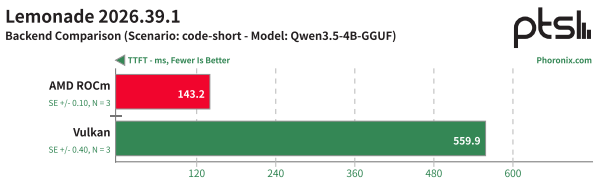
AMD ROCm stack had the benefit of lower latency with the time to first token.





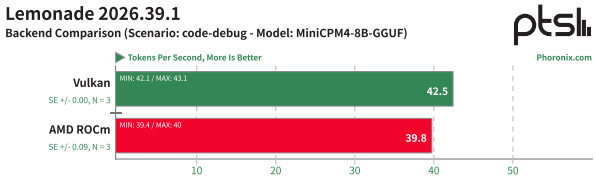
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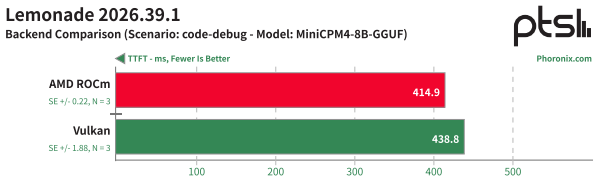


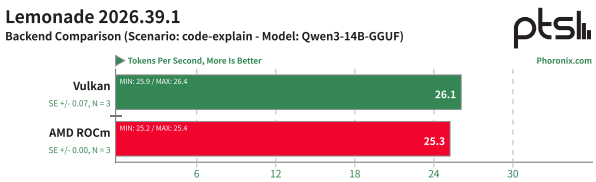


With Qwen3.5 4B, Vulkan still was yielding higher throughput but ROCm was exhibiting quicker times to the first token.

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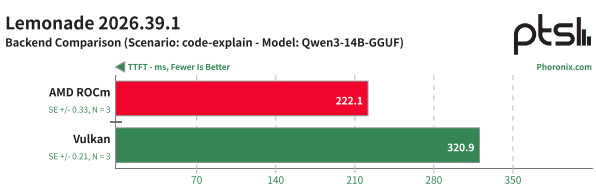
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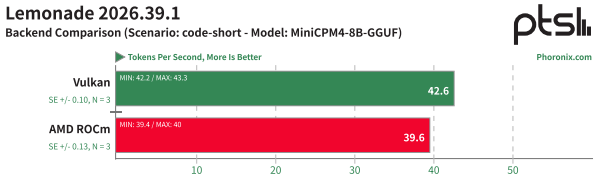
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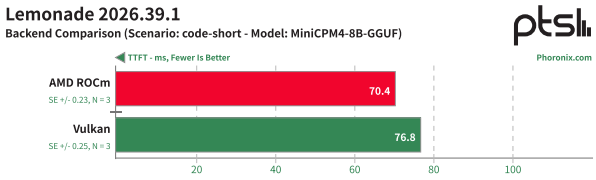
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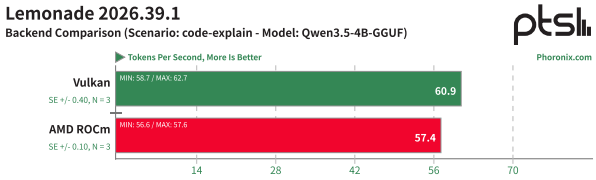
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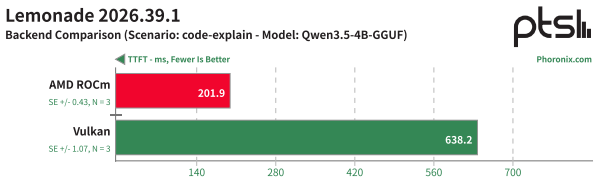

Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: Qwen3-14B-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
AMD ROCm
SE +/- 0.33, N = 3
222.1
Vulkan
SE +/- 0.21, N = 3
320.9
70
140
210
280
350

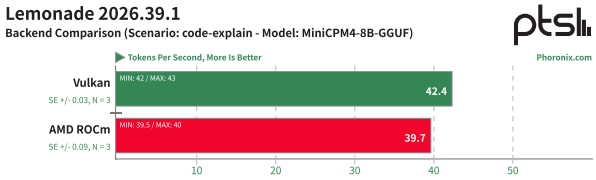

Lemonade 2026.39.1
Backend Comparison (Scenario: code-short - Model: MiniCPM4-8B-GGUF)
Phoronix.com
Tokens Per Second, More Is Better
Vulkan
SE +/- 0.10, N = 3
MIN: 42.2 / MAX: 43.3
42.6
AMD ROCm
SE +/- 0.13, N = 3
MIN: 39.4 / MAX: 40
39.6
10
20
30
40
50

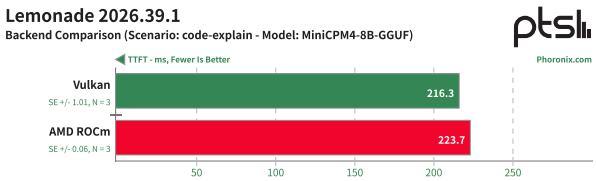
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Lemonade 2026.39.1
Backend Comparison (Scenario: code-short - Model: MiniCPM4-8B-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
AMD ROCm
SE +/- 0.23, N = 3
70.4
Vulkan
SE +/- 0.25, N = 3
76.8
20
40
60
80
100

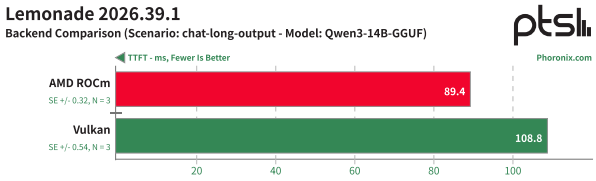

Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: Qwen3.5-4B-GGUF)
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Tokens Per Second, More Is Better
Vulkan
SE +/- 0.40, N = 3
MIN: 58.7 / MAX: 62.7
60.9
AMD ROCm
SE +/- 0.10, N = 3
MIN: 56.6 / MAX: 57.6
57.4
14
28
42
56
70

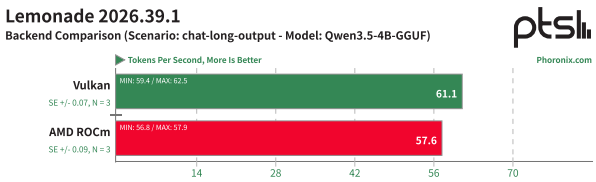

Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: Qwen3.5-4B-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
AMD ROCm
SE +/- 0.43, N = 3
201.9
Vulkan
SE +/- 1.07, N = 3
638.2
140
280
420
560
700


Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: MiniCPM4-8B-GGUF)
Phoronix.com
Tokens Per Second, More Is Better
Vulkan
SE +/- 0.03, N = 3
MIN: 42 / MAX: 43
42.4
AMD ROCm
SE +/- 0.09, N = 3
MIN: 39.5 / MAX: 40
39.7
10
20
30
40
50


Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: MiniCPM4-8B-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
Vulkan
SE +/- 1.01, N = 3
216.3
AMD ROCm
SE +/- 0.06, N = 3
223.7
50
100
150
200
250


Lemonade 2026.39.1
Backend Comparison (Scenario: chat-long-output - Model: Qwen3-14B-GGUF)
Phoronix.com
Tokens Per Second, More Is Better
Vulkan
SE +/- 0.03, N = 3
MIN: 26 / MAX: 26.3
26.1
AMD ROCm
SE +/- 0.03, N = 3
MIN: 25.2 / MAX: 25.5
25.4
6
12
18
24
30

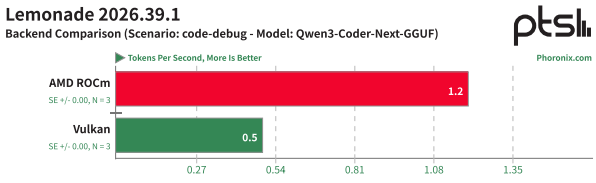

Lemonade 2026.39.1
Backend Comparison (Scenario: chat-long-output - Model: Qwen3-14B-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
AMD ROCm
SE +/- 0.32, N = 3
89.4
Vulkan
SE +/- 0.54, N = 3
108.8
20
40
60
80
100

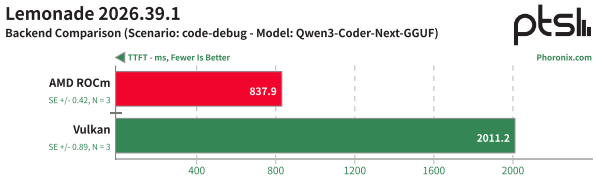




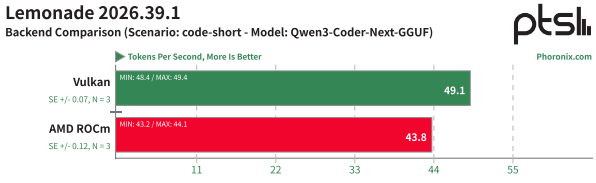
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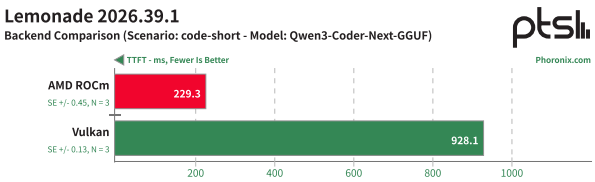
Continuing on across different models and scenarios with the AMD Lemonade local AI server with Llama.cpp, the Vulkan back-end in 2026 was still delivering consistently better performance than ROCm. Typically slightly better but consistent nevertheless. ROCm on the other hand was consistently achieving lower times to the first token over Vulkan, sometimes with very significant margins.

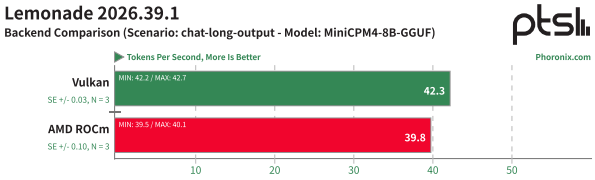




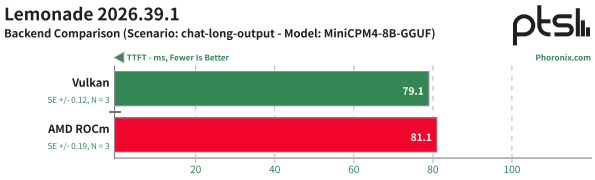
ROCm did pick up a throughput win for code debugging with Qwen3-Coder-Next-GGUF, granted rather slow performance still in this configuration.

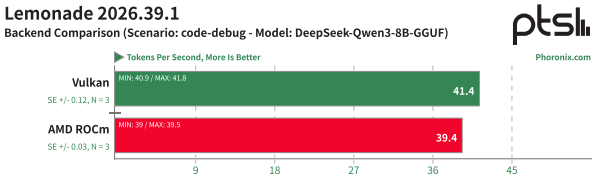


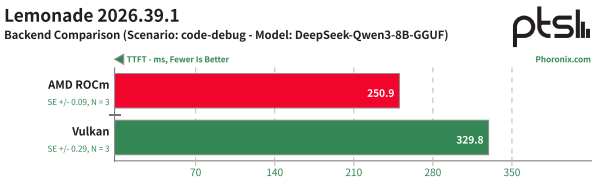

Lemonade 2026.39.1
Backend Comparison (Scenario: code-short - Model: Qwen3-Coder-Next-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
AMD ROCm
SE +/- 0.45, N = 3
229.3
Vulkan
SE +/- 0.13, N = 3
928.1
200
400
600
800
1000

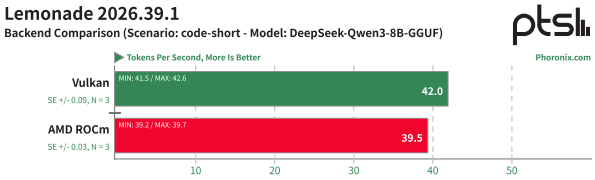

Lemonade 2026.39.1
Backend Comparison (Scenario: chat-long-output - Model: MiniCPM4-8B-GGUF)
Phoronix.com
Tokens Per Second, More Is Better
Vulkan
SE +/- 0.03, N = 3
MIN: 42.2 / MAX: 42.7
42.3
AMD ROCm
SE +/- 0.10, N = 3
MIN: 39.5 / MAX: 40.1
39.8
10
20
30
40
50

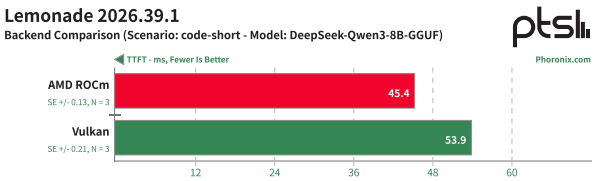
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Lemonade 2026.39.1
Backend Comparison (Scenario: chat-long-output - Model: MiniCPM4-8B-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
Vulkan
SE +/- 0.12, N = 3
79.1
AMD ROCm
SE +/- 0.19, N = 3
81.1
20
40
60
80
100

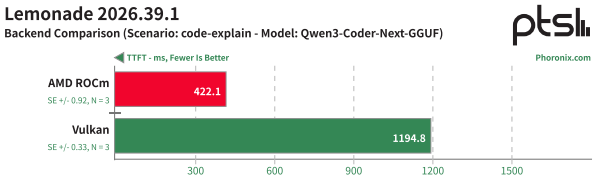

Lemonade 2026.39.1
Backend Comparison (Scenario: code-debug - Model: DeepSeek-Qwen3-8B-GGUF)
Phoronix.com
Tokens Per Second, More Is Better
Vulkan
SE +/- 0.12, N = 3
MIN: 40.9 / MAX: 41.8
41.4
AMD ROCm
SE +/- 0.03, N = 3
MIN: 39 / MAX: 39.5
39.4
9
18
27
36
45

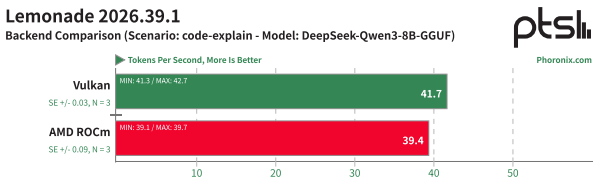

Lemonade 2026.39.1
Backend Comparison (Scenario: code-debug - Model: DeepSeek-Qwen3-8B-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
AMD ROCm
SE +/- 0.09, N = 3
250.9
Vulkan
SE +/- 0.29, N = 3
329.8
70
140
210
280
350

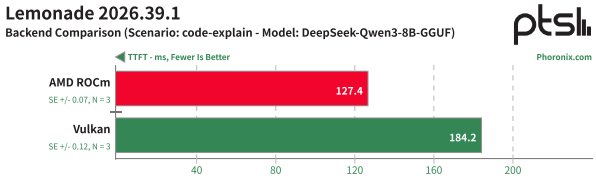

Lemonade 2026.39.1
Backend Comparison (Scenario: code-short - Model: DeepSeek-Qwen3-8B-GGUF)
Phoronix.com
Tokens Per Second, More Is Better
Vulkan
SE +/- 0.09, N = 3
MIN: 41.5 / MAX: 42.6
42.0
AMD ROCm
SE +/- 0.03, N = 3
MIN: 39.2 / MAX: 39.7
39.5
10
20
30
40
50


Lemonade 2026.39.1
Backend Comparison (Scenario: code-short - Model: DeepSeek-Qwen3-8B-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
AMD ROCm
SE +/- 0.13, N = 3
45.4
Vulkan
SE +/- 0.21, N = 3
53.9
12
24
36
48
60

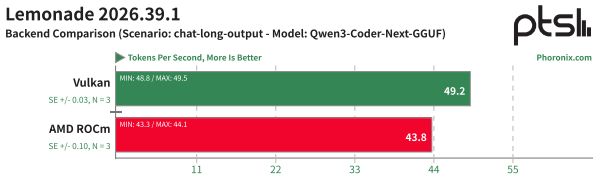

Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: Qwen3-Coder-Next-GGUF)
Phoronix.com
Tokens Per Second, More Is Better
Vulkan
SE +/- 0.07, N = 3
MIN: 48.7 / MAX: 49.3
49.0
AMD ROCm
SE +/- 0.03, N = 3
MIN: 43.2 / MAX: 44
43.7
11
22
33
44
55

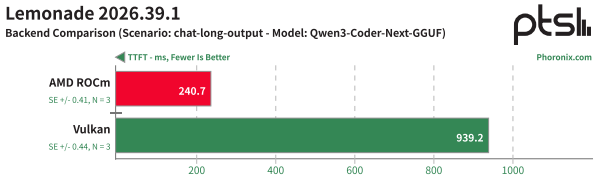

Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: Qwen3-Coder-Next-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
AMD ROCm
SE +/- 0.92, N = 3
422.1
Vulkan
SE +/- 0.33, N = 3
1194.8
300
600
900
1200
1500

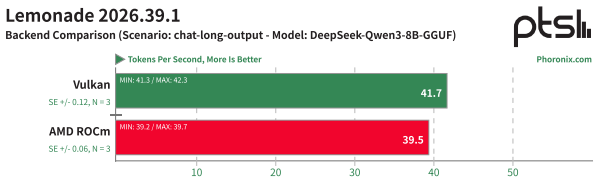


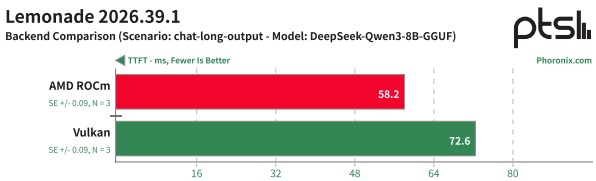


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Page 2 - Strix Halo Benchmarks

Page: 1 2 3 Next Page

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